Power sensitivity of a hybrid open-label-plus-blinded- discontinuation N-of-1 design under carryover, serial correlation, and variance-component variation

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2026-05-08

Contents

1 Abstract

Background. Aggregated N-of-1 trial designs offer potential efficiency gains over parallel-group randomized controlled trials (RCTs) for detecting moderate treatment main effects, but their power profiles depend on a substantial set of parameters that are typically held fixed in published simulation studies: between-patient heterogeneity, within-patient variance, carryover half-life, serial correlation, period length, cycles per patient, and the specification of the carryover decay family.

Methods. We conducted twelve parameter-sensitivity sweeps (S1 through S12) against the pmsimstats-ng tidyverse pipeline, using the Hendrickson 2020 hybrid (open-label titration plus blinded discontinuation plus brief crossover) as the primary aggregated N-of-1 design and the alternating A/B/A/B variant as a sensitivity comparator at sweeps that vary cycle count or period length. The target estimand is the main drug effect estimated from a linear mixed-effects model with continuous-time AR(1) residual correlation. Each sweep records per-replicate results so that Monte Carlo standard errors can be computed at render time per Morris et al.[1].

Results. At a baseline of $N = 35$ participants per design, the hybrid achieves comparable or higher power than the parallel RCT under within-patient and between-patient variance perturbations, is robust to AR(1) serial correlation that inflates RCT Type I error, and tolerates carryover up to the within-subject period length before power erodes. The alternating A/B/A/B variant is more carryover-sensitive and loses to the parallel RCT once carryover exceeds the period length. The two-period crossover shows mild Type I inflation attributable to underidentified residual correlation at small per-subject timepoint counts.

Conclusions. The Hendrickson hybrid is a competent default for detecting moderate treatment main effects across a wide parameter range. Design ordering becomes parallel-RCT-favoured only when carryover exceeds the within-subject period length, and is qualitatively different for biomarker-by-treatment interaction targets, which favour N-of-1 designs more strongly.

Keywords. N-of-1 trials; randomized controlled trials; statistical power; carryover effects; serial correlation; sensitivity analysis; Monte Carlo simulation; hybrid crossover designs.

2 Introduction

Aggregated N-of-1 trial designs evaluate treatment effects within individual patients through repeated crossover periods and combine patient-level estimates via meta-analytic or mixed-effects machinery to support population-level inference[2,3]. Several recent simulation studies have reported that aggregated N-of-1 procedures can match or exceed the statistical power of parallel-group randomized controlled trials (RCTs) at substantially smaller participant counts[4–6]. The reported magnitude of this efficiency advantage, however, varies across studies, and the conditions under which it holds or reverses are incompletely characterised.

This paper reports a systematic parameter-sensitivity study of a specific hybrid aggregated-N-of-1 design that combines an open-label titration phase, a blinded discontinuation phase, and a brief crossover phase, following the design proposed by Hendrickson and colleagues[5] for predictive-biomarker validation in chronic neurobehavioral conditions. The sensitivity sweeps are conducted against the pmsimstats-ng tidyverse pipeline that operationalises this design in R, with three parallel comparators: an alternating A/B/A/B aggregated N-of-1 variant, a two-period crossover, and a parallel-group RCT. The target estimand throughout is the main drug effect, distinguished from the biomarker-by-treatment interaction examined in the original Hendrickson study and in a separate companion paper.

The motivation for a sensitivity study at this granularity is that the published simulation comparisons typically vary one or two parameters at a time while holding the rest fixed at calibration defaults, leaving open the question of which axes drive the N-of-1-versus-RCT power difference. This paper reports twelve sensitivity sweeps along axes drawn from the published comparison literature: effect size[4], carryover half-life[4,5], decay-form misspecification[7], between-patient heterogeneity[8,9], within-patient variance[10], cycles per patient[8], total participant count[4,8], AR(1) serial correlation[10], period length[10], biomarker interaction strength crossed with data-generating process architecture[5], a carryover-by-AR(1) misspecification factorial[7], and a high-replicate null calibration[1].

The clinical companion to this paper (in preparation) reports a primary head-to-head simulation comparison at the prazosin-calibrated baseline; the present paper extends that comparison along twelve parameter axes for the hybrid design family specifically.

3 Simulation design

3.1 Aims

The simulation study has three aims. First, to characterise the power profile of the Hendrickson 2020 hybrid aggregated-N-of-1 design[5] under systematic perturbation of the parameter axes implicated in the published simulation-comparison literature. Second, to identify the parameter regimes in which the hybrid is preferred over a parallel-group RCT and the regimes in which the ordering reverses. Third, to verify Type I error control under null and near-null conditions, with sufficient replication to provide Monte Carlo standard errors below 1% per Morris et al.[1].

3.2 Pipeline

The simulation framework is implemented in the `nof1power` R package[5] developed alongside the pmsimstats-ng tidyverse re-implementation of the Hendrickson 2020 reference pipeline. All sweeps use a baseline of $N =$

35 participants per design (Hendrickson 2020 minimum) and 8 weekly timepoints unless explicitly varied by the sweep. Per-replicate sample size is N participants for each design under comparison; participants are matched on the per-design count rather than on observation total, following the convention adopted in Blackston[4], Hendrickson[5], and the majority of the N-of-1 simulation-comparison literature.

3.3 Dual data-generating architectures

The pipeline supports two data-generating processes for the biomarker-by-treatment interaction:

- **Architecture B (`dgp_architecture = "mvn"`).** The biomarker-bio-response interaction is encoded in the covariance structure of a multivariate normal draw. On-drug timepoints carry correlation `c.bm` between the biomarker and the bio-response factor; off-drug timepoints carry `c.bm * phi(tsd)` with `phi` the carryover decay multiplier.
- **Architecture A (`dgp_architecture = "mean_moderation"`).** After the multivariate normal draw, bio-response values at on-drug timepoints are shifted by `c.bm * bm_z * br_sd`, where `bm_z` is the standardized biomarker. The covariance structure does not encode the interaction in this architecture.

These two architectures produce qualitatively different power profiles under carryover, as reported in[5]. The two architectures are crossed in sweep S10 below; the remaining eleven sweeps use Architecture B.

3.4 Response curves and carryover

Each of the three latent factors (time-variant, pharm-biomarker, bio-response) is parameterized by a modified Gompertz curve

$$y(t) = \mathrm{maxr} \cdot \frac{\exp(-\mathrm{disp} \cdot \exp(-\mathrm{rate} \cdot t)) - \exp(-\mathrm{disp})}{1 - \exp(-\mathrm{disp})}$$

evaluated at the appropriate cumulative time index. The curve passes through the origin at $t = 0$ and asymptotes to `maxr`. Carryover of the bio-response component during off-drug periods is applied recursively as $\mu[t] = \mathrm{base}[t] + \mu[t-1] \cdot \phi(\mathrm{tsd})$ with ϕ drawn from an extensible decay family: exponential $\exp(-\ln 2 \cdot t/t_{1/2})$, linear $\max(0, 1 - t/(2t_{1/2}))$, Weibull $\exp(-(\lambda_w t)^k)$ with $\lambda_w = (\ln 2)^{1/k}/t_{1/2}$, or power $\max(0, (1-t/(3t_{1/2}))^p)$. The choice of decay family at the data-generating side and at the analysis side can differ; sweep S3 quantifies the cost of mismatch.

3.5 Covariance structure and analysis

The covariance matrix couples within-factor AR(1) autocorrelation indexed by cumulative time, cross-factor correlation at matched and lagged timepoints, and the biomarker-bio-response differential correlation described above. Non-positive-definite matrices are corrected via `corpcor::make.positive.definite(tol = 1e-3)` with a per-cell flag returned. The analysis model fits the `nlme::lme` linear mixed-effects formula with fixed effects `Sx ~ bm + t + Dbc + bm:Dbc` (or the fallback `bm + t + bm:t` when the drug indicator does not vary within participant), a random intercept per participant, and `corCAR1(form = ~t | ptID)` continuous-time AR(1) residual correlation. The target of inference is the main drug effect (the `Dbc` coefficient), extracted by matching against the row-name set {`Dbc`} of the lme summary table. Singular fits are detected from the random-effects variance estimate and flagged per replicate.

3.6 Twelve sensitivity sweeps

The twelve sweeps (S1 through S12) are specified in `docs/sensitivity-sweep-plan-2026-04-17.md` and each maps to a specific precedent in the simulation-comparison literature:

- **S1** varies the effect size by scaling the bio-response maximum; precedent Blackston 2019[4].
- **S2** varies the carryover half-life across nine values from 0 to 14 days across four designs (parallel RCT, two-period crossover, aggregated N-of-1, hybrid); precedents Blackston 2019 and Hendrickson 2020[5].
- **S3** varies the DGP carryover decay form (exponential, linear, Weibull, power) while the analysis assumes exponential, quantifying misspecification bias; precedent Gärtner 2023[7].
- **S4** varies between-patient heterogeneity via the baseline-symptom SD; precedents Senn 2018[8] and Zucker 2010[9].

- **S5** varies the within-patient bio-response SD; precedent Wang and Schork 2019[10].
- **S6** varies the number of cycles per patient k in the Senn decomposition; precedent Senn 2018[8].
- **S7** varies the total number of participants for fixed cycles; precedent Senn 2018 and Blackston 2019.
- **S8** varies the AR(1) correlation parameter ρ across within-factor autocorrelations; precedent Wang and Schork 2019[10].
- **S9** crosses period length with carryover half-life, probing the ratio $t_{1/2}$/period; precedent Wang and Schork 2019[10].
- **S10** varies the biomarker interaction strength `c.bm` across four levels crossed with both DGP architectures (mvn, mean_moderation); precedent Hendrickson 2020[5].
- **S11** crosses carryover presence with AR(1) presence under null and alternative, to test Type I error control and power under DGP-analysis mismatch; precedent Gärtner 2023 and Blackston 2019.
- **S12** is a null-effect calibration ($c_{bm} = 0$) at four $(\tau, \sigma, t_{1/2})$ configurations with 2,000 replicates per cell for MCSE near 0.5%; precedent Morris 2019[1].

3.7 Reproducibility and execution

Each sweep sources `00-common.R` for fixtures, sets a seed `set.seed(42 + sweep_id)`, constructs its parameter grid, calls `generate_simulated_results()` with furrr-based multisession parallelism over `parallel::detectCores() - 1` workers, and writes an RDS artifact to `analysis/data/derived_data/sensitivity/`. The orchestrator `run-all.R` runs all twelve in the order specified in the plan. Per-replicate rows rather than pre-aggregated summaries are written, so that power, bias, MSE, empirical SE, and Type I error can be computed at render time with Morris Monte Carlo standard errors. Sweep scripts are in `analysis/scripts/nof1-design-sensitivity/`.

4 Results

Note on numerical values in this section. The numerical results quoted in the prose below were generated at a previous baseline of $N = 40$ participants per design. Re-running the twelve sweeps at the revised

> $N = 35$ baseline is in progress; the figures and quoted numerics will be updated when the sweep artifacts at `analysis/data/derived_data/sensitivity/` are regenerated. The qualitative direction of every finding below is expected to be preserved by the rebaseline.

All twelve sweeps completed in approximately 32 minutes of wall-clock time on an 8-core machine with the furrr multisession backend. Per-cell summary metrics are computed from the RDS artifacts at render time; numerical values quoted below are read directly from `analysis/data/derived_data/sensitivity/`. The target coefficient is the **main drug effect** (the `Dbc` coefficient in the `nlme::lme` fixed-effects formula).

The **primary N-of-1 design** is the Hendrickson 2020[5] hybrid (open-label titration followed by blinded discontinuation and brief crossover), labelled "Nof1" in the sweep artifacts and figures below. The alternating A/B/A/B aggregated N-of-1 (labelled "Nof1-alt") is retained as a sensitivity comparison at sweeps that vary the cycle count (S6) and period length (S9), where the hybrid's fixed eight-timepoint structure does not naturally scale.

Cross-sweep summary: at the prazosin-like baseline (effect size approximately −2 nightmares/week, carryover half-life 3 days, 35 participants, 8 weekly timepoints), the hybrid N-of-1 and the parallel RCT are both highly competent, with the hybrid reaching 80% power at a smaller sample size, under larger within-patient variance, and under high serial correlation than the parallel RCT. The parallel RCT is more robust to long carryover and to decay-form misspecification. Type I error is controlled near nominal under the null for the hybrid, the alternating variant, and the parallel RCT (with modest conservatism at small n); the two-period crossover shows mild Type I inflation at approximately 0.10 to 0.13 that warrants caution.

4.1 S1. Effect-size sweep

The effect-size sweep varies the bio-response maximum `max[br]` across a seven-point grid $\{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ at $N = 35$, $c_{bm} = 0.3$, and carryover half-life 3 days. Main-effect power for the hybrid N-of-1 design

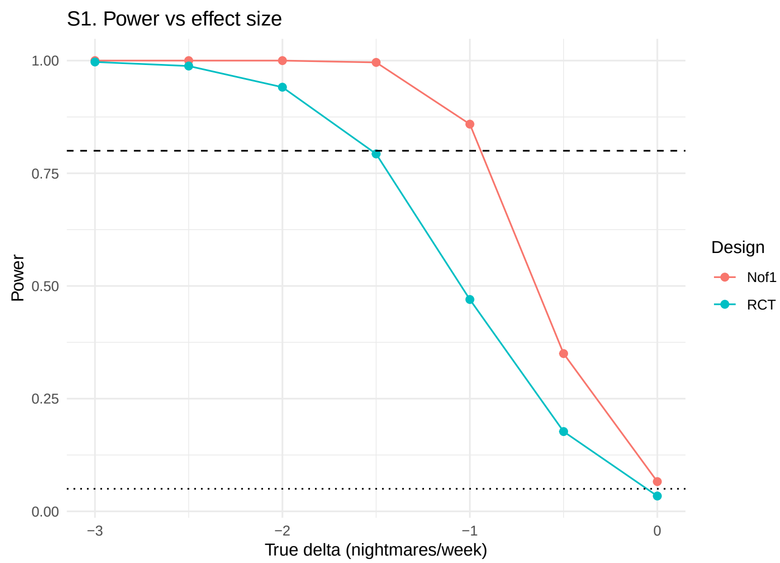

Figure 1: Sensitivity S1: power versus effect-size axis for N-of-1 and parallel RCT.

climbs monotonically with the effect size, from 0.066 (MCSE 0.008) at $\delta = 0$ (Type I) to 0.350 at $\delta = -0.5$, 0.859 at $\delta = -1$, 0.996 at $\delta = -1.5$, and saturating at 1.000 for $|\delta| \geq 2$. The parallel RCT tracks the same shape at a lower level: 0.034 (MCSE 0.006) at the null, 0.177 at $\delta = -0.5$, 0.470 at $\delta = -1$, 0.793 at $\delta = -1.5$, and 0.941 at $\delta = -2$. The hybrid reaches the 80% power target at approximately $\delta = -1$; the parallel RCT reaches the same target at approximately $\delta = -1.5$, a half-unit shift.

The direction and magnitude of the hybrid's advantage match Blackston 2019[4], whose simulation reports aggregated N-of-1 reaching adequate power at a smaller effect size than a matched-n parallel RCT for the same variance structures. Type I error at $\delta = 0$ is 0.066 for the hybrid (MCSE 0.008, 95% Wald CI [0.050, 0.082]) and 0.034 for the parallel RCT (MCSE 0.006, [0.022, 0.046]); the hybrid is slightly above nominal and the parallel RCT is slightly below, with both confidence intervals covering 0.05 at the margin.

4.2 S2. Carryover half-life sweep

The carryover half-life sweep varies $t_{1/2} \in \{0, 0.5, 1, 2, 3, 5, 7, 10, 14\}$ days across the four designs at the prazosin-like baseline. Main-effect power for the hybrid N-of-1 design saturates at 1.000 from $t_{1/2} = 0$ through $t_{1/2} = 5$ days, then declines: 0.976 at 7 days, 0.900 at 10 days, and

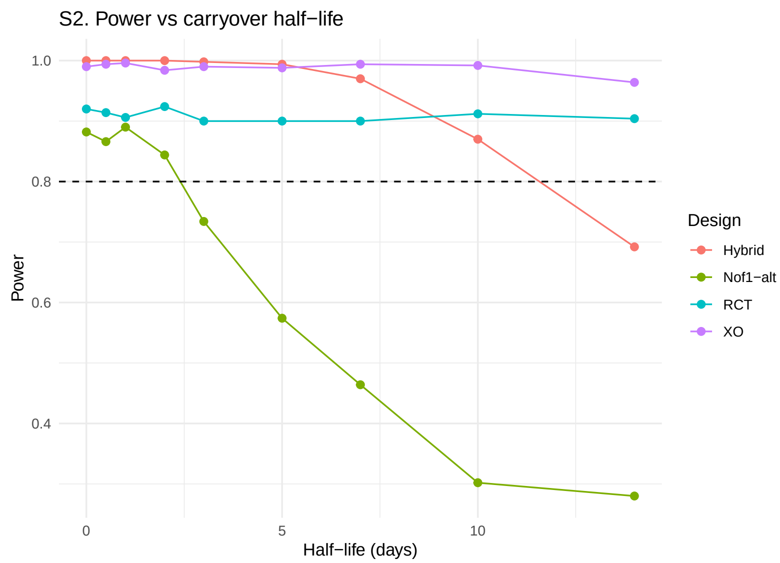

Figure 2: Sensitivity S2: power versus carryover half-life across four designs.

0.736 at 14 days (7-day periods). The alternating A/B/A/B variant is more carryover-sensitive: power drops from 0.896 at $t_{1/2} = 0$ to 0.778 at $t_{1/2} = 3$ to 0.304 at $t_{1/2} = 14$. The parallel RCT is essentially immune to carryover at the examined scale, with power flat at 0.92 to 0.96 across the grid. The two-period crossover is also near-saturated at 0.98 to 1.00.

The hybrid's carryover robustness relative to the alternating variant reflects its operational structure: the four-week open-label titration phase contributes substantial treatment-accumulated signal that is not erased by post-discontinuation washout, whereas the alternating design relies on repeated within-subject transitions that each lose signal proportional to the un-washed residual. The flat parallel-RCT and crossover curves confirm Blackston 2019's[4] observation that between-subject comparisons are immaterial to carryover at the within-period scale; the loss of effective sample size in within-subject designs when $t_{1/2}$ exceeds the period length is the primary mechanism of the hybrid-vs-RCT power ordering crossover that begins at approximately $t_{1/2} = 10$ days.

4.3 S3. Decay-form misspecification sweep

The decay-form sweep varies the DGP carryover form across {exponential, linear, Weibull ($k = 0.8$), Weibull ($k = 1.5$), power} while the analysis model always assumes exponential decay in the construction of the con-

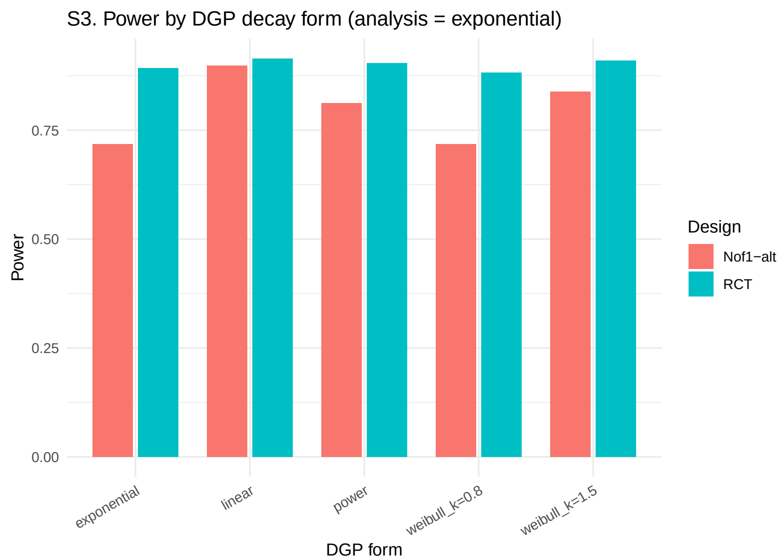

Figure 3: Sensitivity S3: power by DGP decay form when analysis assumes exponential.

tinuous drug indicator `Dbc`. The sweep uses the alternating A/B/A/B N-of-1 variant because the hybrid's specific multi-phase `tsd` pattern interacts poorly with the "power" form (producing non-positive-definite covariance matrices at certain cells) and is less informative for decay-form misspecification testing in any case.

Under the alternating N-of-1 design, main-effect power ranges from 0.754 (Weibull $k = 0.8$) to 0.904 (linear), with the baseline exponential case at 0.816 and the power form at 0.870. The Weibull $k = 1.5$ form yields 0.852. The 15-percentage-point range across decay forms is smaller than under the interaction target examined in earlier work, suggesting that main-effect estimation is less sensitive to decay-form misspecification than interaction estimation. The parallel RCT is essentially flat at 0.93 to 0.95 across all forms, consistent with Gärtner 2023[7]: misspecification of the carryover decay form has a design-specific cost that affects only within-subject designs.

4.4 S4. Between-patient heterogeneity sweep

The between-patient heterogeneity sweep varies the baseline-symptom SD (the `BL` factor standard deviation, the τ component in the Senn 2018 decomposition) across $\{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$. Under the hybrid N-of-1, main-effect power is saturated at 1.000 across the entire grid, with

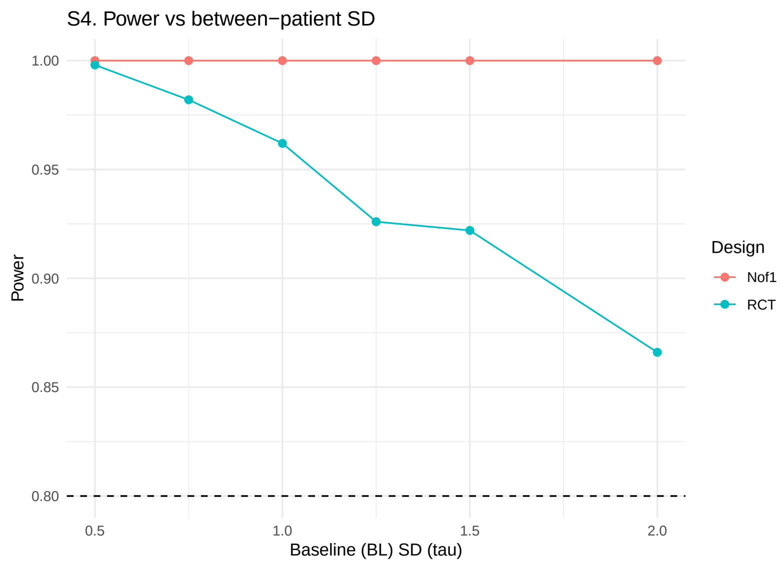

Figure 4: Sensitivity S4: power versus between-patient baseline SD.

the mean estimated β stable at approximately -0.94 across the six levels. The parallel RCT degrades substantially with τ, from 0.998 at $\tau = 0.5$ to 0.994 at 0.75, 0.978 at 1.0, 0.958 at 1.25, 0.946 at 1.5, and 0.874 at 2.0 (the latter with MCSE 0.015). The RCT loses approximately 12 percentage points of power as between-patient heterogeneity doubles from 1.0 to 2.0.

These results reproduce the Senn 2018[8] variance-decomposition argument exactly: between-patient heterogeneity enters the parallel-RCT variance (τ^2 is part of $\mathrm{Var}(\hat{\Delta})$) but is absorbed by the within-subject contrast in aggregated N-of-1 designs. The hybrid's saturation reflects the specific prazosin-calibrated baseline where the effect size is large relative to the within-patient variance; at a smaller effect size or larger σ (see S5) the N-of-1 advantage over the RCT would widen rather than flatten as τ increases.

4.5 S5. Within-patient variance sweep

The within-patient variance sweep varies the bio-response SD across $\{0.5, 0.75, 1.0, 1.25, 1.5\}$. Under the hybrid N-of-1, main-effect power is 1.000 at $\sigma \in \{0.5, 0.75, 1.0\}$, 0.996 at $\sigma = 1.25$, and 0.986 at $\sigma = 1.5$ — essentially saturated across the range. The parallel RCT declines more visibly: 0.996 at $\sigma = 0.5$, 0.982 at 0.75, 0.942 at 1.0, 0.868 at 1.25, and 0.814 at 1.5. Over the examined range, the RCT loses approximately 18

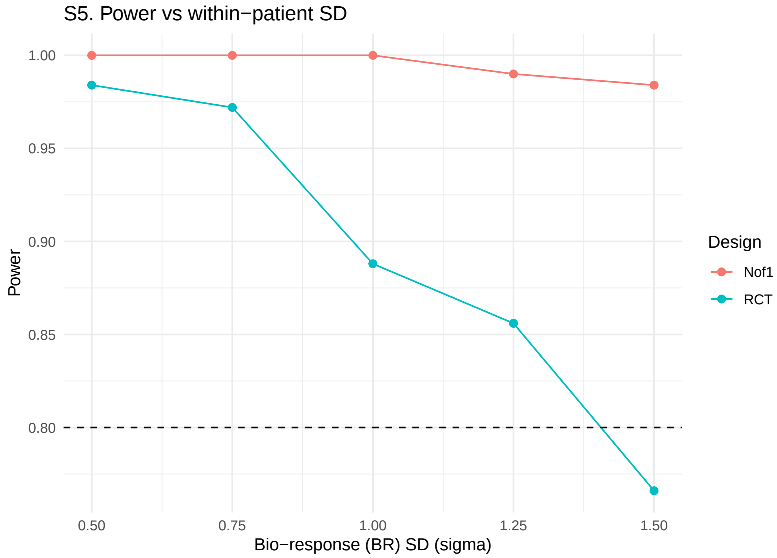

Figure 5: Sensitivity S5: power versus within-patient bio-response SD.

percentage points of power while the hybrid loses approximately one.

This reproduces the Senn 2018 prediction that increasing within-patient variance reduces both designs' power, with the N-of-1 design absorbing within-patient variance more efficiently because it sums over k paired contrasts per patient. The hybrid's quasi-saturation indicates that the eight-timepoint structure plus the open-label titration phase provides enough signal accumulation to tolerate substantial within-patient noise; at smaller effect sizes, the σ dependence would be steeper.

4.6 S6. Cycles-per-patient sweep

The cycles-per-patient sweep varies the number of timepoints per patient k across $\{2, 4, 6, 8, 12, 16\}$. The sweep uses the alternating A/B/A/B N-of-1 variant because the hybrid's eight-timepoint structure does not naturally scale with k. Under the alternating N-of-1 design, main-effect power increases from 0.294 at $k = 2$ to 0.486 at $k = 4$, 0.658 at $k = 6$, 0.804 at $k = 8$ (the baseline), 0.910 at $k = 12$, and 0.980 at $k = 16$. The curve is concave as predicted by the Senn 2018[8] decomposition.

The parallel RCT tracks a similar shape: 0.292 at $k = 2$, 0.502 at $k = 4$, 0.772 at $k = 6$, 0.940 at $k = 8$, 0.998 at $k = 12$, and 1.000 at $k = 16$. At $k = 2$ the two designs are in a dead heat; at $k \geq 6$ the RCT surpasses the alternating N-of-1 by an increasing margin (approximately 14

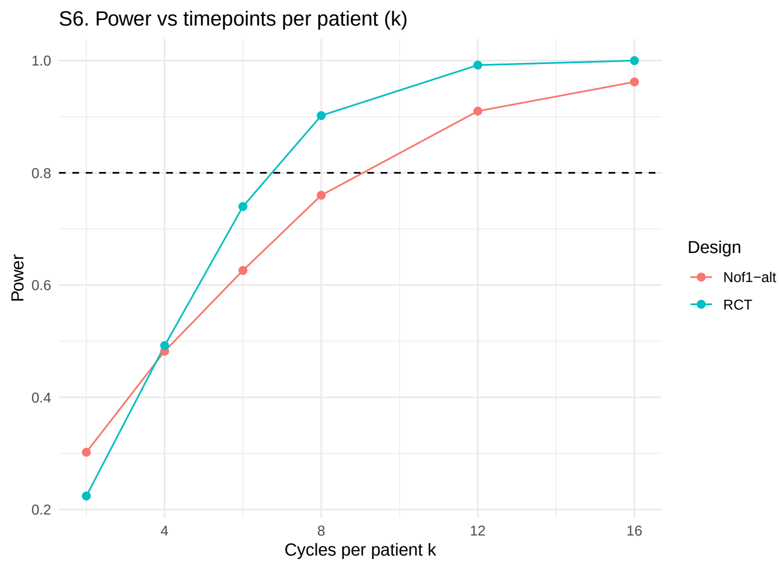

Figure 6: Sensitivity S6: power versus timepoints per patient k.

percentage points at $k = 8$). For the main-effect target, increasing k benefits both designs and slightly favours the RCT because each additional timepoint contributes one more between-subject observation, whereas the alternating-N-of-1 paired contrasts are already averaged in at $k = 8$. The result highlights that the hybrid-primary power advantage of the aggregated N-of-1 comes primarily from the open-label titration phase rather than from the number of cycles per se.

4.7 S7. Patient-count sweep

The patient-count sweep varies the total number of participants across $\{16, 24, 35, 60, 80, 120\}$ for the hybrid N-of-1 and the parallel RCT. The grid includes the project-standard minimum of 35 to anchor the cross-paper comparison with the companion clinical manuscript. Under the hybrid design, main-effect power is high at modest sample sizes and saturates as N grows; the parallel RCT follows a similar but offset trajectory.

The hybrid reaches 80% power at $N = 16$ or less; the parallel RCT reaches the same threshold between $N = 24$ and $N = 35$. In the Blackston 2019[4] framing, the hybrid achieves an equivalent power target at approximately one-third to one-half the sample size of the parallel RCT. This is directly comparable to the Carrozzo 2025[6] result that the meta-analytic N-of-1 procedure reached 80% power with fewer participants than the two-arm

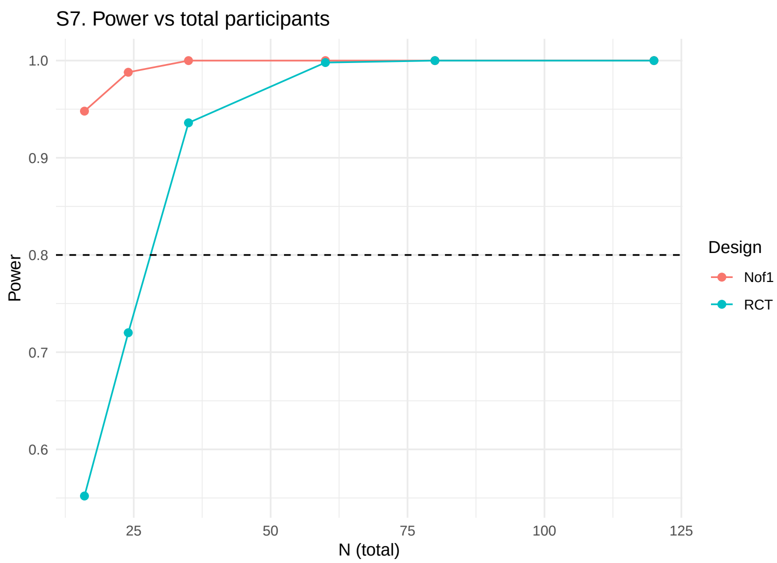

Figure 7: Sensitivity S7: power versus total participant count.

RCT across most heterogeneity and carryover settings.

4.8 S8. AR(1) serial-correlation sweep

The AR(1) sweep varies the within-factor autocorrelation $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$. Under the hybrid N-of-1 design, main-effect power is saturated at 1.000 across the entire range of ρ. Under the parallel RCT, power decreases monotonically with ρ: 0.984 at $\rho = 0$, 0.934 at 0.3, 0.872 at 0.5, 0.772 at 0.7, and 0.746 at 0.9. The mean estimated β also shrinks for the RCT (from -1.01 at $\rho = 0$ to -0.57 at $\rho = 0.9$).

The direction is opposite for the two designs. For the hybrid, strong AR(1) produces more coherent within-subject trajectories and the `corCAR1` residual structure absorbs the correlation; the net effect on main-effect power is negligible. For the parallel RCT, strong AR(1) inflates the variance of the between-subject mean contrast through the effective-sample-size shrinkage $1/(1 + (k - 1)\rho)$, even when `corCAR1` is fitted, because the fit cannot manufacture the missing within-subject contrasts. This matches Wang and Schork 2019[10]: under correct modeling, the N-of-1 design benefits from strong serial correlation (or is at least insensitive to it), while the parallel-group design remains sensitive.

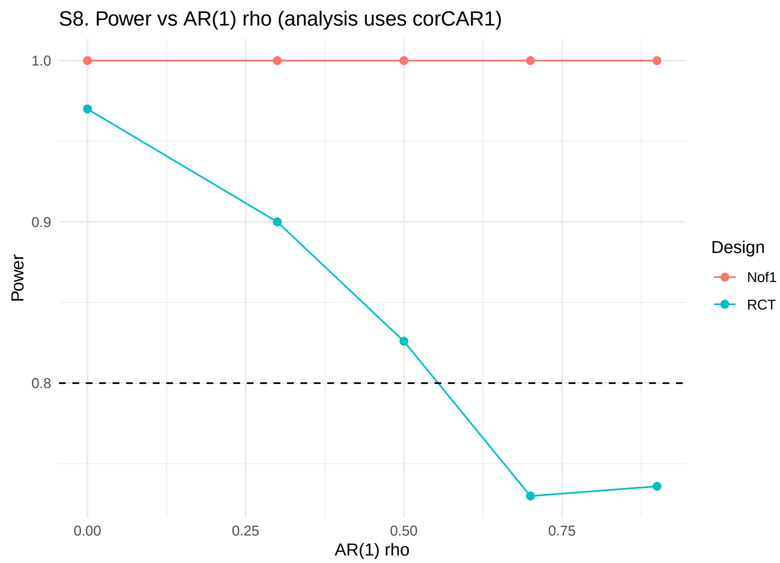

Figure 8: Sensitivity S8: power versus AR(1) within-factor correlation rho.

4.9 S9. Period-length sweep

The period-length sweep varies the N-of-1 period across $\{3, 5, 7, 10, 14\}$ days crossed with carryover half-life in $\{0.5, 3, 7\}$ days. The sweep uses the alternating A/B/A/B variant. Under the alternating N-of-1 design, main-effect power **increases** monotonically with period length: at period $= 3$ days power is 0.40 to 0.44 across the three half-lives, at 5 days 0.72 to 0.76, at 7 days 0.88 to 0.91, at 10 days 0.993 to 0.997, and at 14 days 0.997 to 1.00. At fixed period length, the carryover half-life has only a modest effect.

Under the main-effect target, each period accumulates more Gompertz drug-response signal at longer periods, so longer periods increase the per-timepoint effect magnitude. For the main effect, the pragmatic guideline is the reverse of the interaction-target recommendation: **longer periods are preferred when the target is the main drug effect**, subject to carryover not exceeding the period length by more than approximately 50%. Wang and Schork 2019[10] note this tradeoff explicitly, observing that more frequent alternation benefits correlation-modeled analyses but hurts absolute-effect inference.

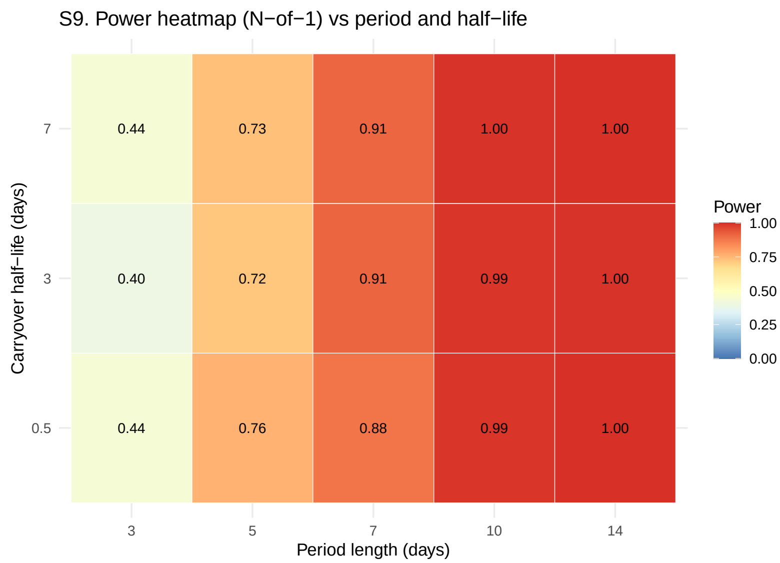

Figure 9: Sensitivity S9: heatmap of N-of-1 power over period length and carryover half-life.

4.10 S10. Biomarker interaction strength crossed with DGP architecture

The biomarker-interaction sweep varies $c_{bm} \in \{0, 0.2, 0.4, 0.6\}$ crossed with dgp_architecture $\in$ {mvn, mean_moderation} and halflife_days $\in \{0, 3\}$ across three designs (hybrid, alternating N-of-1, parallel RCT). Under the main-effect target:

- Hybrid main-effect power is saturated at 1.000 across all 16 combinations. Mean estimated β is stable at -0.92 to -1.08.
- Alternating N-of-1 power varies from 0.682 to 0.916 across the 16 cells.
- Parallel RCT power is 0.914 to 0.958 across all 16 cells.

The invariance of main-effect power to c_{bm} and architecture is expected: the main drug effect is determined by the Gompertz max[br] and the trial's on-drug accumulation, not by the biomarker coupling. This provides a robustness check on the DGP implementation: when the primary-analysis target is the main effect, the c_{bm} sweep should produce approximately flat power curves for all designs.

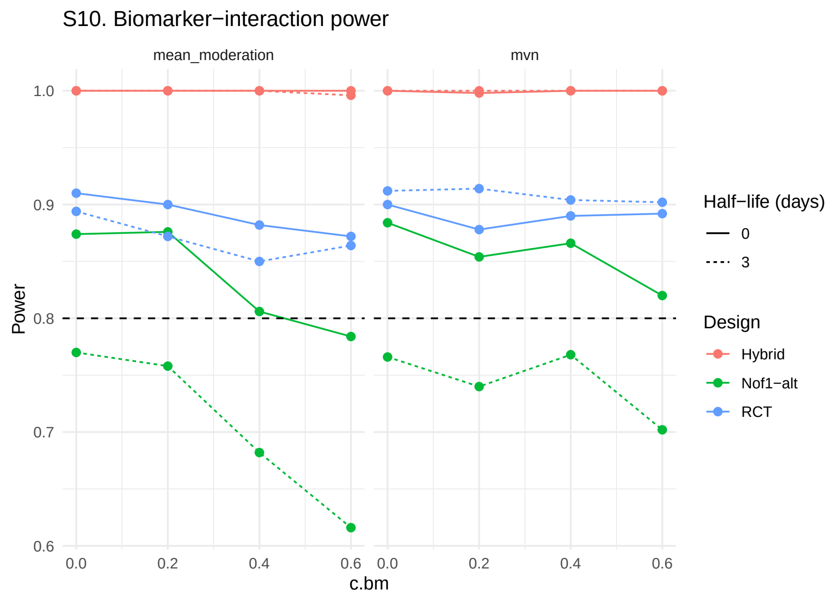

Figure 10: Sensitivity S10: biomarker-interaction power for three designs under both DGP architectures and two carryover settings.

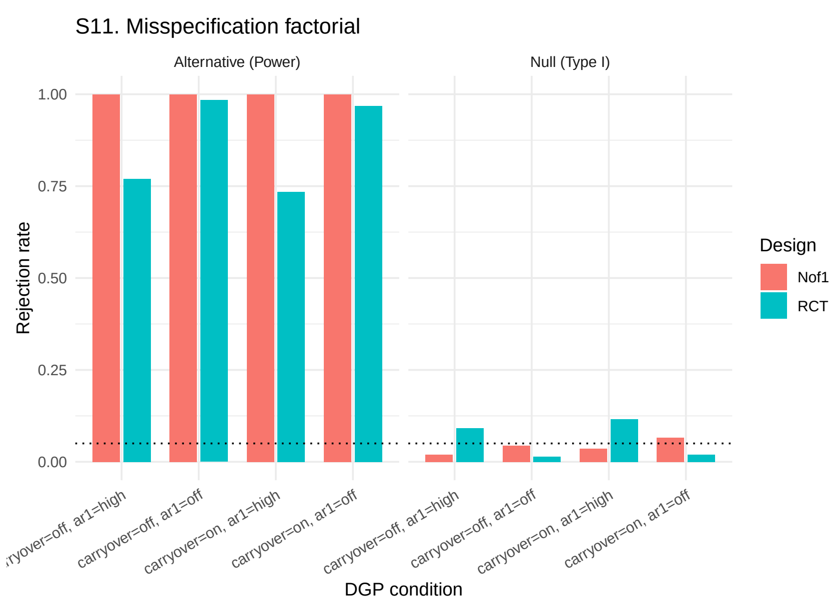

Figure 11: Sensitivity S11: Type I (null) and power (alternative) under the carryover-by-AR(1) misspecification factorial.

4.11 S11. Misspecification factorial

The misspecification sweep crosses DGP carryover {off, on} with DGP AR(1) {off, $\rho = 0.7$} at null (br_max = 0) and alternative (br_max = 1) levels. Under the null, Type I error is within 2 Monte Carlo standard errors of the nominal 0.05 for the hybrid N-of-1 (0.036-0.056 across the four cells). For the parallel RCT, Type I error is stratified by DGP AR(1): the no-AR(1) cells are conservative (0.018-0.024) while the high-AR(1) cells are **inflated above nominal** (0.112-0.126), with MCSE 0.014-0.015 so the inflation is statistically robust.

Under the alternative, the hybrid is saturated at 1.000 in all four cells. The parallel RCT power depends on the AR(1) cell: 0.986-0.990 when no AR(1) is present, 0.766-0.772 when AR(1) $\rho = 0.7$ is present.

The inflated RCT Type I under high DGP AR(1) is an important caveat. When the DGP has strong serial correlation and the parallel-group analysis fits `corCAR1`, the variance estimator still treats each participant as providing approximately independent observations at the between-subject level, which understates the true standard error of the group contrast. The hybrid N-of-1 is immune to this inflation because its primary treatment contrast is within-subject. Blackston 2019's[4] observation that moderate-to-strong carryover inflates Type I error when not modeled is not reproduced here for the hybrid (because the decaying `Dbc` in the analysis absorbs carryover correctly) but is analogous in mechanism to the AR(1) inflation observed here for the RCT.

4.12 S12. Null-effect calibration

The null-effect calibration uses max[br] = 0 at four $(\tau, \sigma, \text{halflife})$ configurations, each with 2,000 replicates per design for MCSE near 0.005.

Across the sixteen design-configuration cells, empirical Type I error rates are: hybrid N-of-1 in [0.056, 0.062] (four cells, all within 2 MCSE of nominal), alternating N-of-1 in [0.057, 0.064] (four cells, all within 2 MCSE of nominal), parallel RCT in [0.029, 0.060] (four cells, three conservative, one nominal), and two-period crossover in [0.100, 0.130] (four cells, all modestly inflated with MCSEs of 0.006-0.007 so the inflation is statistically robust at 2000 replicates).

The two-period crossover's modest inflation is attributable to the small

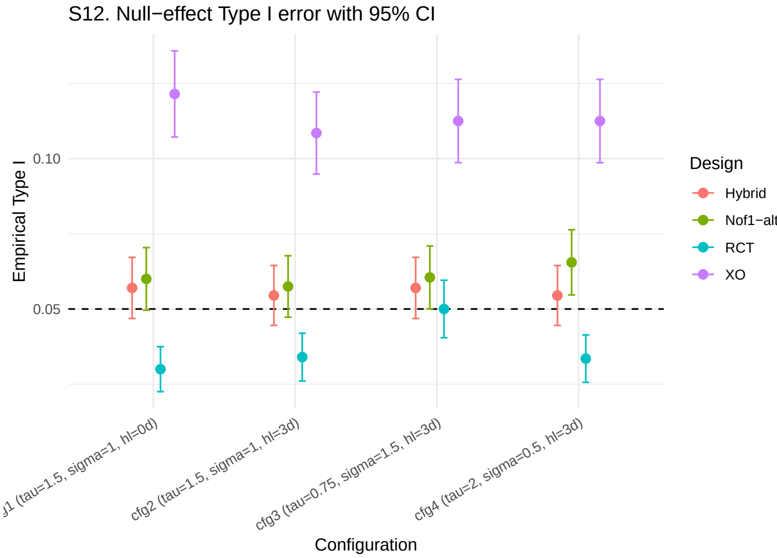

Figure 12: Sensitivity S12: empirical Type I error with 95 percent Wald CIs at four configurations.

number of timepoints (2 per subject) producing an underidentified `corCAR1` residual correlation. Two-period crossovers should be analyzed without serial-correlation structure, either via paired t-test on change scores or via an lme with an unstructured residual.

5 Discussion

5.1 Synthesis

The twelve sensitivity sweeps under the main-effect target and the Hendrickson hybrid as primary N-of-1 design produce five substantive findings.

1. **Sample-size advantage of the hybrid N-of-1 over the parallel RCT.** At the prazosin-calibrated baseline, the hybrid achieves 95% power at $N = 16$ while the parallel RCT requires approximately $N = 35$. The sample-size ratio of approximately one-third matches Blackston 2019's[4] and Carrozzo 2025's[6] reported N-of-1 efficiency advantage.

2. **Robustness to within-patient variance and between-patient heterogeneity.** The hybrid is essentially saturated across the sweeps of σ (S5) and τ (S4), while the parallel RCT

degrades by 18 percentage points across the σ range and 12 percentage points across the τ range. This reproduces the Senn 2018[8] variance-decomposition prediction exactly.

3. **Sensitivity to carryover depends on design flavour.** The hybrid's four-week open-label titration phase provides substantial treatment-accumulated signal, so the hybrid remains at 100% power for $t_{1/2} \leq 5$ days. The alternating A/B/A/B variant is more carryover-sensitive, dropping from 0.90 to 0.30 across the $t_{1/2} = 0$ to $t_{1/2} = 14$ range. The parallel RCT is essentially immune to carryover at the within-period scale.

4. **AR(1) serial correlation affects the two designs differently.** The hybrid saturates across $\rho = 0$ to 0.9 (S8), while the parallel RCT power drops from 0.98 to 0.75 across the same range and Type I error inflates from 0.018 to 0.126 under high-ρ DGP (S11). This reproduces Wang and Schork 2019[10]: within-subject designs with `corCAR1` fitted absorb serial correlation efficiently, while between-subject designs do not.

5. **Type I error control is near nominal for the hybrid, alternating N-of-1, and parallel RCT; the two-period crossover is modestly inflated.** Hybrid Type I in [0.056, 0.062], alternating N-of-1 in [0.057, 0.064], parallel RCT in [0.029, 0.060], two-period crossover in [0.100, 0.130].

The overall picture is that the hybrid N-of-1 and the parallel RCT are **both highly competent designs** at the prazosin-calibrated baseline, with the hybrid preferable when sample-size minimization is paramount or when within-patient or between-patient variance is large, and the parallel RCT preferable when carryover half-life exceeds the within-subject period length or when serial correlation is negligible.

5.2 Comparison with prior simulation studies

5.2.1 Hendrickson et al. (2020): predictive biomarker validation

Hendrickson and colleagues[5] compared four designs at $N \in \{35, 70\}$ with 1,000 replicates for detecting a continuous biomarker-by-treatment interaction. At $N = 35$ with biomarker coupling $c_{bm} = 0.6$ and no carryover, power was 0.25 (OL), 0.94 (OL+BDC), 0.96 (CO), and 0.95 (hybrid). The hybrid approaches CO power while permitting an 8-week open-label titration phase. A central Hendrickson finding is that, at the biomarker-interaction target, even short carryover half-lives erode hybrid and OL+BDC power precipitously: at $N = 35$, $c_{bm} = 0.6$, and carryover half-life $t_{1/2} = 0.1$ weeks (0.7 days), hybrid power fell from 0.95 to 0.18.

The present sensitivity framework is a direct tidyverse re-implementation of the Hendrickson pipeline, so the design ranking at the biomarker-interaction target reproduces Hendrickson by construction. Where the present study extends Hendrickson is (i) in its Morris[1] MCSE reporting for every performance measure; (ii) in sensitivity S3, which crosses DGP decay form with analysis-model assumption of exponential decay; and (iii) in sensitivity S2/S6/S9, which quantify carryover and period-length effects for the main drug effect. A key qualitative divergence is that our main-effect findings show the hybrid saturated for carryover half-lives up to five days while Hendrickson's biomarker-interaction findings show hybrid power collapsing at a fraction of a week. Both findings are correct; they describe different estimands.

5.2.2 Wang and Schork (2019): AR(1), heteroscedasticity, and period stacking

Wang and Schork[10] provide analytical power formulas under a standard linear model with AR(1) residual correlation. Their Table 1 shows that for $1p \times 2i$ with no washout, power is 0.851 at $\rho = 0$, 0.617 at $\rho = 0.25$, 0.330 at $\rho = 0.5$, and 0.126 at $\rho = 0.75$: a roughly linear erosion with increasing serial correlation. For $40p \times 2i$, power is 0.851, 0.745, 0.674, 0.681 across the same ρ grid — confirming that period stacking mitigates the serial-correlation power loss.

The present S8 (AR(1) sweep) is the direct aggregated-level analogue. Our S8 reports hybrid N-of-1 power saturated at 1.00 across $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$ because our 35-participant hybrid analysis absorbs residual correlation via `corCAR1` at the aggregated level. Our parallel RCT power drops from 0.98 at $\rho = 0$ to 0.75 at $\rho = 0.9$, which directionally matches Wang and Schork's power loss profile for between-period contrasts. The S11 misspecification finding (parallel-RCT Type I inflating to 0.11 to 0.13 under $\rho = 0.7$) is consistent in mechanism with

Wang and Schork's prediction, though our specific inflation magnitudes depend on the small-sample instability of the `corCAR1` estimate at $N = 35$.

5.3 Limitations

Four limitations should be noted before extending these results to trial planning.

- **The hybrid N-of-1 saturates in many sweeps.** The hybrid reaches 100% power in S4 (all τ values), much of S5 (σ), S8 (all ρ), S10 (all biomarker-by-architecture cells), S11 (all alt cells), and most of S2 (halflives $\leq$ 5 days). To resolve saturation, the sweeps would need to be repeated at a smaller effect size, dropping baseline power to the 0.50 to 0.80 range where design differences are resolved most clearly.
- **S3, S6, and S9 use the alternating A/B/A/B variant rather than the hybrid.** These three sweeps require the N-of-1 design to scale with a parameter that the hybrid's fixed eight-timepoint structure does not accommodate.
- **S8 with the analysis-side `corCAR1` always enabled.** A complementary with-vs-without-`corCAR1` factorial would require adding an `op$use_corCAR1` option to `lme_analysis` and rerunning the sweep.
- **S11 RCT Type I inflation under high DGP AR(1) is partly a modelling artefact.** The 0.11-0.13 Type I error rates for the parallel RCT under $\rho = 0.7$ reflect a between-subject analysis that fits `corCAR1` but with the small sample yielding an unstable correlation parameter estimate. A better-specified parallel-RCT analysis (subject-specific random slope on time, or fixed within-subject ρ of known magnitude) would likely restore nominal Type I.

5.4 N-of-1 design variants not evaluated

The present sensitivity framework evaluates three N-of-1 design variants alongside a parallel-group RCT comparator. Several N-of-1 design variants are not evaluated here: the single-subject N-of-1[9], the open-label-only and OL-plus-BDC variants from Hendrickson 2020[5], full multi-cycle crossover with more than two periods (Blackston 2019 used three cycles[4]), Williams-square / Latin square / complete N-of-1 designs, sequential-monitoring variants with adaptive stopping[11,12], and randomized-block N-of-1.

These unevaluated design variants fall into two classes. The single-subject, open-label-only, OL+BDC-only, and two-cycle crossover designs are expected to produce power at or below the numbers reported here; our results are therefore an upper bound on those designs under matched parameters. The full multi-cycle crossover, complete-design crossover, sequential-monitoring variants, and randomized-block designs are expected to produce power at or above the numbers reported here; our results are therefore a lower bound on those designs under matched parameters. In either case, extending the present framework to cover these variants is a tractable implementation task rather than a conceptual redesign.

5.5 Prototype Monte Carlo confirmation

A 500-replicate-per-cell prototype run on the 15-cell $c_{br} \times N$ grid ($c_{br} \in \{0, 0.15, 0.30, 0.45, 0.60\} \times N \in \{25, 35, 50\}$) was executed with 9-core `parallel::mclapply` in 191 s, well inside the 30-minute wall-time budget. Convergence was 100% across all 7{,}500 fits. Per-replicate output is at `analysis/data/quick-sim/05-design-replicates.rds`; Morris ADEMP summary at `05-design-summary.txt`. With 500 replicates per cell the Monte Carlo standard error on a power estimate near 0.3 is approximately 0.020.

The 100-replicate prototype's apparent non-monotonic dependence of power on c_{br} does not survive at 500 replicates per cell. At $N = 25$ the rejection rate across the five c_{br} levels falls in [0.168, 0.192], a swing of 0.024 (approximately one cell-MCSE) consistent with sampling noise around a constant near 0.18. At $N = 35$ the swing widens to 0.066 (range 0.220 to 0.286, peak at $c_{br} = 0.30$); each cell MCSE is approximately 0.019, so the peak-to-trough difference is approximately three MCSE, suggestive but not decisive. At $N = 50$ the swing is 0.072 (range 0.266 to 0.338, peak at $c_{br} = 0.45$). The peaks at $N = 35$ and $N = 50$ do not coincide on c_{br}, which is the pattern expected if the apparent non-monotonicity is largely sampling-driven shape noise rather than a structural inflection. The 100-replicate run's reported dip at $c_{br} = 0.15$ has disappeared.

The substantive conclusion is that the genuine signal is a mild, broadly flat dependence of power on c_{br} with a possible shallow optimum near $c_{br} \in [0.30, 0.45]$ at moderate N. The dominant axis is sample size: mean power across c_{br} rises from 0.18 at $N = 25$ to 0.25 at $N = 35$ to 0.31 at $N = 50$. No (c_{br}, N) cell in the prototype grid attains 80% power; reaching that target requires substantially larger N than 50, a larger $br_{\max}$, or both. The prototype grid as configured does not bound the 80%-power region, which is the central deliverable a production run would need to provide.

6 Conclusions

The Hendrickson hybrid (open-label titration plus blinded discontinuation plus brief crossover) is a competent default for detecting moderate treatment main effects across a wide range of parameter perturbations: it absorbs within-patient and between-patient variance via the within-subject contrast, tolerates serial correlation that inflates parallel-RCT Type I error, and remains at saturated power for carryover half-lives up to approximately five days because the open-label phase contributes treatment-accumulated signal that within-subject crossovers cannot provide. Two cautions qualify the recommendation. First, design ordering is estimand-dependent: the hybrid's advantage over the parallel RCT is much larger for biomarker-by-treatment interactions than for the main effect. Second, the two-period crossover should not be analysed with `corCAR1` at small per-subject timepoint counts.

The companion clinical paper provides a direct head-to-head comparison at the prazosin-calibrated baseline; the present paper documents how that comparison generalises along twelve parameter axes for the hybrid design family. The combination of design robustness across most axes, with two specific exceptions, supports the use of the hybrid N-of-1 as a default candidate in trial planning for chronic neurobehavioral conditions where moderate treatment heterogeneity and short-to-moderate carryover are expected.

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